

Principles of Economics

Class 1

Introduction, the capitalist revolution,
technology, population and growth

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Introduction: What is economics?

An example of an economic problem: the capitalist revolution

The economist's tools: models, data, narratives

The Malthusian model

The technology-choice model

Other related issues

1. Introduction: What is economics?

Course contents – Part 1 (weeks 1 to 7)

Question	Tool	Unit ¹	Week
<i>What questions arise from the capitalist revolution?</i> Wealth creation, environmental impact, inequalities	Measuring output and inequality	Mi1	1
<i>Why did the capitalist revolution happen when and where it did?</i> Pursuit of profit and competition → innovation and productivity	Innovation rents, production function	Mi2	1
<i>How to make decisions in non-strategic settings?</i> Economic models: doing the best you can	Constrained optimization	Mi3	2
<i>How to make decisions when interacting with others?</i> Self-interest, altruism and cooperation	Game theory, Nash equilibrium, experiments	Mi4	3
<i>Why do firms exist and how are conflicts of interest resolved?</i> Wage setting	Principal-agent model	Mi6	4
<i>How do firms interact with their customers?</i> Price-setting firms	Elasticity, consumer and producer surplus	Mi7	5
<i>How does a market with many buyers and sellers work?</i> Price-taking firms	Supply and demand, market equilibrium	Mi8	6
<i>When are markets inefficient and what can the government do?</i> Externalities and public goods	Private and social costs, Pigouvian taxes	Mi10	7

¹ Units follow CORE's *The Economy 2.0*: Mi = Microeconomics, Ma = Macroeconomics.

Course contents – Part 2 (weeks 8 to 14)

Question	Tool	Unit ¹	Week
<i>How does the labour market of the whole economy work? Wage setting, price setting and unemployment.</i>	The labour market model	Ma1	8
<i>How do credit, money and banks expand (and sometimes limit) opportunities?</i>	Borrowing, lending and investment	Mi9 / Ma6	9
<i>Why does the economy go through booms and busts and how does that affect employment?</i>	The business cycle	Ma3	10
<i>How does the government use its power to raise revenue or spend to improve the economy?</i>	Fiscal policy and unemployment	Ma5	11
<i>How do central banks use their power to create money to stabilize the economy?</i>	Monetary policy and inflation	Ma4	12
<i>What determines technological progress, employment and living standards in the long run?</i>	Economic growth	Ma9	13
<i>How are income disparity, wealth and power measured in the economy and what is done to change them?</i>	Inequality measures, redistribution policy	Ma2	14
<i>How is the environment being destroyed and what can we do about it?</i>	Climate change and pollution	Ma8	14

¹ Units follow CORE's *The Economy 2.0*: Mi = Microeconomics, Ma = Macroeconomics.

What is the most pressing problem economists should be addressing? (2024–25)



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“An Inquiry into the Nature and Causes of the Wealth of Nations”: production, consumption and accumulation of wealth (exchangeable goods and services).

Adam Smith (1723–1790)

“Economics is the **study of mankind** in the ordinary business of life. It examines that part of individual and social action which **is most closely connected with the attainment** of wellbeing and the use of the material requisites for its achievement”: wealth as a means to wellbeing.



Alfred Marshall (1842–1924)



"Economics as the science which studies human behaviour as a **relationship between ends and scarce means** which have alternative uses."

Lionel Robbins (1898–1984)



"Economics is a **relation between people** disguised as a relation between things."

Karl Marx (1818–1883)



Against "We may abstract from **entrepreneurs** and treat productive services as being exchanged directly for one another."

Léon Walras (1834–1910)



That is, economics as a theory of decision-making with no **free lunch**.

Milton Friedman (1912–2006)



That is, economics is a **theory of interactive decision-making** – game theory.

John Nash (1928–2015) y John von Neumann (1903–1957)



... or the relationship with **nature** and how to preserve it.

Elinor Ostrom (1933–2012)

► Definition in CORE Econ:

*The study of how people **interact** with each other and with the natural environment to provide themselves with their **livelihoods**, and how this **changes over time**.*

► It deals with:

- How we **acquire** our **livelihood**: food, clothing, shelter, free time.
- How we **interact**: buyers and sellers, employees and employers, citizens and officials, families.
- How we interact with the **natural environment**: from breathing to extracting raw materials.
- How all of this **changes** over **time**.

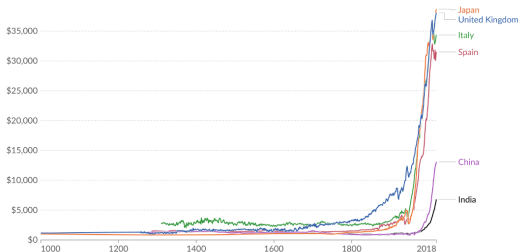
2. An example of an economic problem: the capitalist revolution

Our first question

Rapid growth in average living standards since 1700

History's hockey stick: Worldwide historical real gross domestic product per capita, 1000 to 2018

Unit 1 'The capitalist revolution: prosperity, inequality, and planetary limits' in The CORE Team, The Economy 2.0 Microeconomics. Available at: <https://tinyco.re/19274920> [Figure 1.1]



Data source: Maddison Project Database (2018)

tinyco.re/19274920 | Powered by ourworldindata.org

Note: The units of measurement is 2011 US dollar which is used to compare Purchasing Power Parity and GDP across countries over time. CC-BY-ND-NC

Source: Maddison Project Database (2018). Chart adapted from **CORE Econ** / Our World in Data.

Explore other periods and countries: <https://tinyco.re/3290463>

- Flat until the **18th century** and very rapid thereafter:

- United Kingdom $\approx$ 1700;
- Japan / Italy $\approx$ 1870
- China / India $\approx$ 1960

- “Hockey stick” growth



- What has happened in the last 200 years?
- What is so special about it?

- ▶ **GDP (Gross Domestic Product)**: value of final goods and services produced in the country in a period.
- ▶ **“Domestic”**: produced within the territory, whatever the producer’s nationality.
- ▶ To calculate it:
 - ▶ Count final goods and services and their prices.
 - ▶ **Nominal GDP**: at current-year prices:

$$\text{GDP}_t = \sum_{i=1}^n p_{i,t} q_{i,t}$$

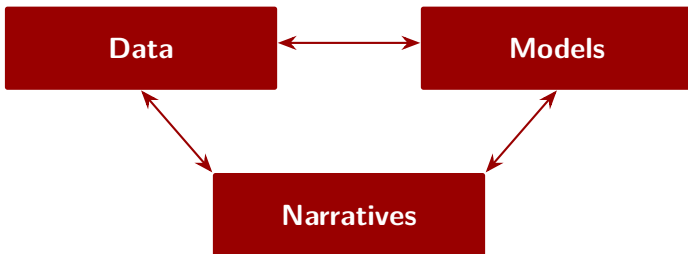
- ▶ Other ways of measuring GDP: week 10.
- ▶ GDP has limitations:
 - ▶ Poor measure of the informal economy and of wellbeing.
 - ▶ Comparisons over time and across countries are tricky.

3. The economist's tools: models, data, narratives

The economist's toolbox

How do economists make sense of the world?

- ▶ Combining **three components**:



- ▶ Various **types of data**:
 - ▶ **Common examples**: GDP, GDP per capita, inflation, sovereign debt, budget revenues, public spending, socio-demographic information, financial asset prices, etc.
- ▶ Less obvious data in some **fields**:
 - ▶ Economics of education: students' grades.
 - ▶ Political economy: election results.
 - ▶ Health economics: hospital patients.
 - ▶ Labour economics: wages.
 - ▶ Environmental economics: emissions.
- ▶ **Diverse sources**:
 - ▶ Direct observation, surveys, administrative records.
 - ▶ Various frequencies: daily, monthly, quarterly, annual.
 - ▶ Different levels of aggregation: municipalities, regions, countries.
- ▶ **New types** from new technologies:
 - ▶ Social media (X/Twitter), news analysis, etc.

- ▶ The economy: decisions and interactions of millions of people.
- ▶ Models simplify: the essential, without the details.
- ▶ How agents act, and what emerges from their interactions.
- ▶ **Equilibrium**: no agent has an incentive to deviate.
- ▶ They show how outcomes respond to external changes (*shocks*).

Ceteris paribus: “all else equal”. The effect of one variable, the rest fixed.

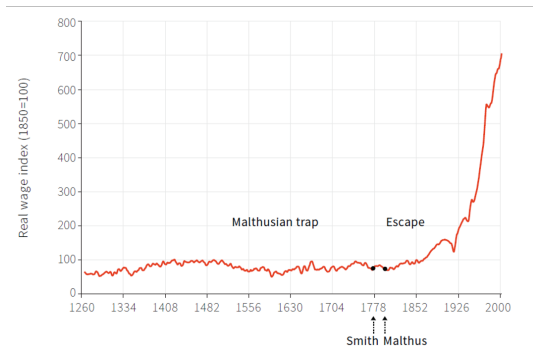
Borges, *On Exactitude in Science* (video).

What is a **good** model?

Four characteristics:

- ▶ Clear: helps us understand something important.
- ▶ Useful: helps improve how the economy works.
- ▶ Accurate: predictions consistent with the evidence.
- ▶ Communicative: shows where we agree or disagree.

- ▶ **Real wages:** key to GDP per capita.
- ▶ **Constant** 1260–1850: the **Malthusian trap**.
- ▶ From 1850, **rapid rise**: $\times 7$ by 2000.
- ▶ The **great escape**.
- ▶ What explains this evolution?



4. The Malthusian model

- ▶ First, the **600 years of stagnation, 1260–1850**:
 - ▶ Useful here: an **economic model**.
 - ▶ Ours: the “**Malthusian trap**” . .
 - ▶ **Intuition**:

Technological advances → improvements in living standards → population growth → the improvements in living standards are diluted.

- ▶ Key: the interaction between **production** and **demographics**.

► **Food production function:**

- Y : quantity of food produced (output).
- X : quantity of labour and land (inputs).

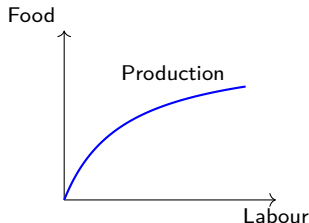
► **Production function:**

$$Y = f(X)$$

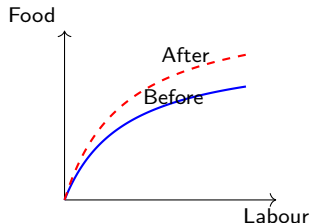
► **We will assume:**

Mathematical notation	Mathematical meaning	Economic meaning
$f(0) = 0$	The function passes through $(0,0)$.	No inputs, no output.
$f(X) > 0$, si $X > 0$	The function is positive if $X > 0$.	Positive inputs, positive output.
$\frac{dY}{dX} > 0$	The function is increasing.	Production increases with inputs.
$\frac{d^2Y}{dX^2} < 0$	The function is concave.	Each extra input adds less (diminishing returns).

- ▶ Food as a function of labour (population), with fixed land.
- ▶ Diminishing returns: each extra worker produces less.
- ▶ Technological improvement: the curve shifts up.

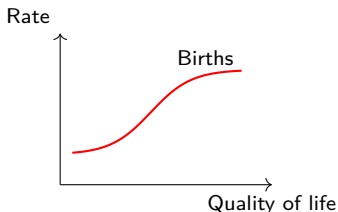


Without technological change

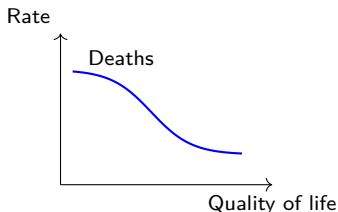


Technological improvement

- ▶ What happens when more food is produced?
- ▶ Quality of life rises.
- ▶ For simplicity, quality of life = food.
- ▶ So population grows, through two effects:



Effect on birth rate



Effect on death rate

And if population grows, what happens to quality of life?

Grain (output) from farmers (labour), with the production function:

$$\text{Average output} = \frac{\text{Total output}}{\text{No. of workers}}$$



$$\text{At A: } \frac{500,000}{800} = 625$$

$$\text{At B: } \frac{732,000}{1,600} = 458$$

► More workers, ever smaller gains in output.

⇒ **Diminishing marginal returns.**

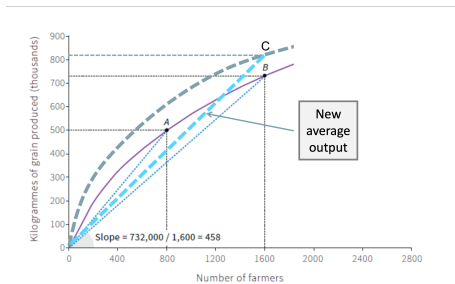
Grain (output) from farmers (labour), with the production function:



$$\text{Average output} = \frac{\text{Total output}}{\text{No. of workers}}$$

- ▶ At B (equilibrium).
- ▶ Technology improves (shock).
- ▶ What happens to:
 - ▶ Output?
 - ▶ And average output?

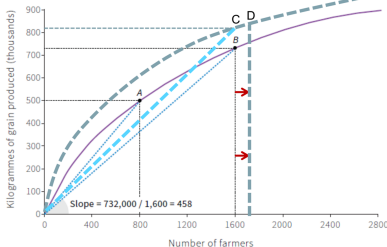
When technology improves:



$$\text{At C: } \frac{820,000}{1600} = 513$$

- ▶ Average output rises, $B \rightarrow C$.
Why? Same workers, more output, thanks to technology.
 $\Rightarrow$ **New, higher average output.**
- ▶ And now, the population?

When population increases:



- ▶ More population, more farmers.
- ▶ Total output rises, average output falls.

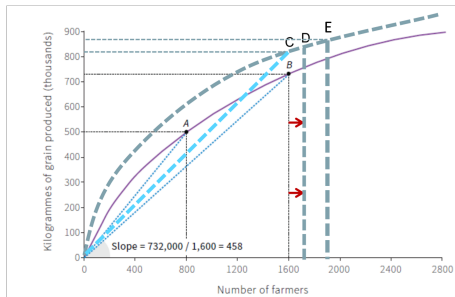
⇒ **Diminishing marginal returns.**

- ▶ At D:

$$\frac{1,024,000}{2000} = 512$$

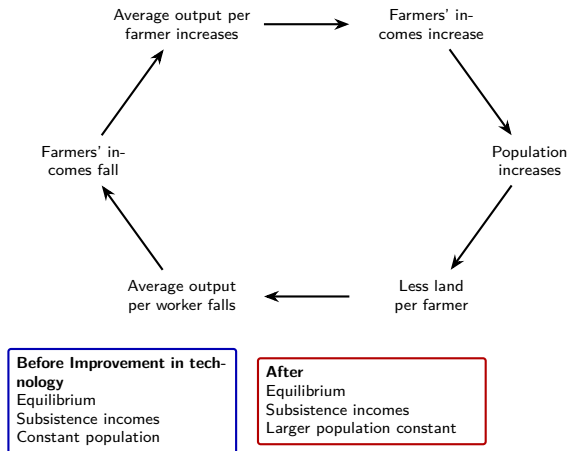
- ▶ Are we better off than at B?
- ▶ More total output, average output barely moves.

If technology does not change, when will population stop growing?



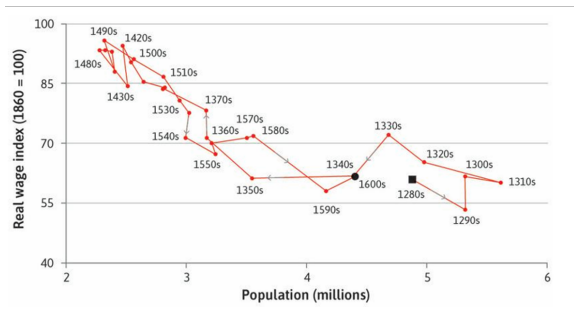
- ▶ Population grows while births > deaths.
- ▶ Average output at E = Average output at B = 458
- ▶ Long run: stagnation at **subsistence**, average output = 458.
- ▶ Equilibrium: **births = deaths** $\Rightarrow$ population stops growing.
- ▶ Technology raises output; quality of life unchanged.
- ▶ Population growth cancels the technological gains.

Result of the Malthusian model



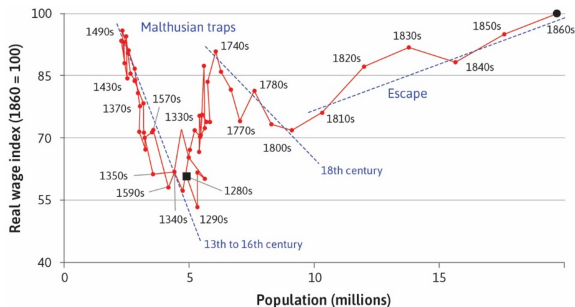
► Key result:
Population changes keep subsistence living standards, despite tech. improvement.

Wage index and population:



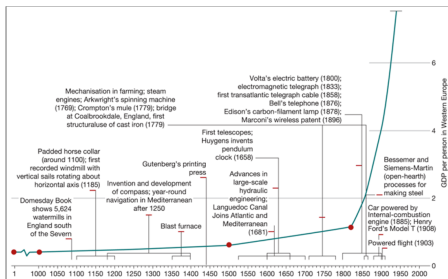
- ▶ Population falls, wages rise.
- ▶ The **negative relationship** is clear.
- ▶ Between **1280–1600**, the **model fits reality!**

- ▶ A good explanation of the stagnation of 1280–1600;
- ▶ does it also explain the great escape?
- ▶ The same chart, now including the 19th century:



- ▶ The negative relationship disappears!
- ▶ The Malthusian model **does not explain** the 19th century.

- ▶ The model is not bad: something changed.
- ▶ What began in Great Britain around 1800?



- ▶ The Industrial Revolution: innovation at an unprecedented rate from the mid-18th century.
- ▶ So fast that population could no longer cancel the gains in living standards.

5. The technology-choice model

- ▶ Unit costs of the inputs:
 - ▶ 1 unit of labour: $w = 10 \text{ €}$
 - ▶ 1 tonne of coal: $r = 20 \text{ €}$
- ▶ Total cost of 100m of cloth:

$$C = wL + rK$$

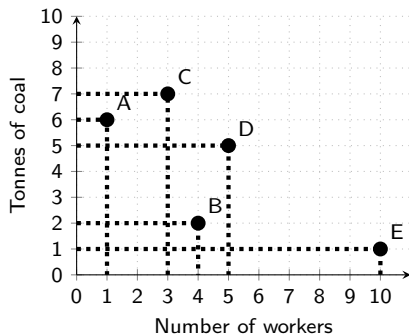
Technology	Labour (L)	Capital (K)	Total cost
A	1	6	130€
B	4	2	80€
C	3	7	170€
D	5	5	150€
E	10	1	120€

- ▶ Which technology does the firm choose?
 - ▶ B is the least costly $\Rightarrow$ lowest iso-cost.
 - ▶ C and D are dominated by A and B $\Rightarrow$ never chosen.
- ▶ Wage-capital ratio (slope): labour relatively cheap $\Rightarrow$ labour-intensive technologies

$$\frac{w}{r} = \frac{10}{20} = 0.5$$

The Great Escape: the role of technology

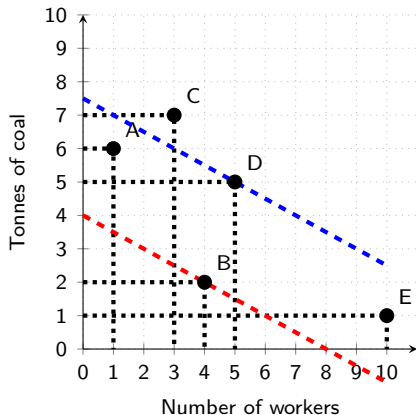
- ▶ Combinations of labour L and capital K to produce 100m of cloth.
- ▶ Some technologies (A) dominate others (C): fewer resources.
- ▶ The firm picks the cheapest technology.
- ▶ C is more expensive than A.
- ▶ But between A and E, which is more expensive?



Technology	L	K	Output
A	1	6	100
B	4	2	100
C	3	7	100
D	5	5	100
E	10	1	100

→ More capital-intensive

→ More labour-intensive



- ▶ **Iso-cost line:** input combinations with the same total cost.
- ▶ With given input prices, **total cost:**

$$C = wL + rK \Leftrightarrow K = -\frac{w}{r}L + \frac{C}{r}$$

where $\frac{w}{r}$ is the slope = wage-capital ratio

- ▶ **Farther from the origin,** more expensive: D costs more than B.

Why did the Industrial Revolution happen in England and not in France?

- ▶ In England the wage / cost-of-capital ratio soared from the early 17th century:
 - ▶ A rise in P , a fall in R , or both.
- ▶ Hence more capital-intensive technologies in England.
- ▶ And a strong incentive to innovate.

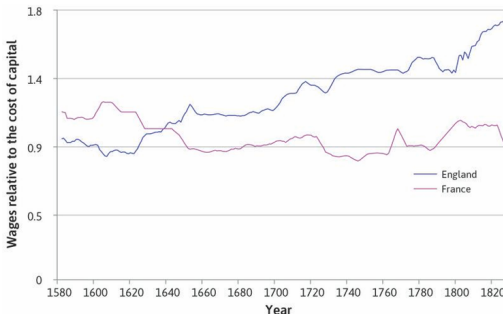


Figure 2.11 Wages relative to the cost of capital goods (late sixteenth to the early nineteenth century).

Page 138 in Robert C. Allen, 2008, *The British Industrial Revolution in Global Perspective*, Cambridge: Cambridge University Press.

- ▶ Coal keeps its price; the wage rises to 40€.

Technology	A	B	C	D	E
Labour	1	4	3	5	10
Capital	6	2	7	5	1
Initial cost	130	80	170	150	120
Final cost	160	200	260	300	420

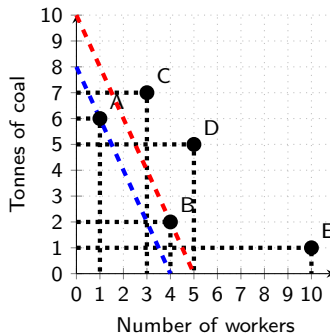
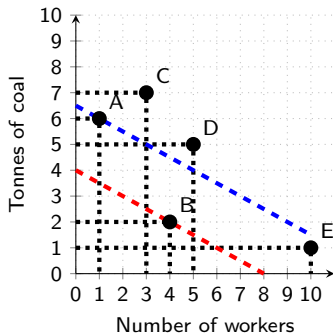
Unit cost	Initial	Final
Capital	20	20
Labour	10	40
w/r	-0.5	-2

Larger (more negative) wage-capital ratio:
capital-intensive technologies are favoured.

- ▶ Which technology now?
 - ▶ A is now the cheapest, for the same output.
 - ▶ D and C, never chosen: dominated by B and A.

The Great Escape: using our model

Iso-cost lines through points A-B with $\{w = 10\text{€}, r = 20\text{€}\}$ and with $\{w = 40\text{€}, r = 20\text{€}\}$



- ▶ Dearer labour **changes the slope of the iso-cost**:
 - ▶ Steeper: capital-intensive technologies become more advantageous.
- ▶ A now lies on an iso-cost closer to the origin:
 - ▶ Hence a lower cost than B.
 - ▶ Cheaper, same output.
- ▶ So, we switch from B to A!

6. Other related issues

The Great Escape: creative destruction

Joseph Schumpeter (1883–1950)

- ▶ **Creative destruction (Joseph Schumpeter, 1883–1950):** new technologies displace the old ones, and the firms that fail to adapt.
- ▶ Creative because bankruptcy frees labour and capital for new combinations.
- ▶ Is it that simple?
 - ▶ Not always: mismatches, and the existing stock of labour and capital.
 - ▶ And possibly large social and environmental costs.



- ▶ Naïve to take input prices as the only factor behind the Great Escape.
- ▶ Alternative explanations:
 - ▶ The scientific revolution in Europe and the Enlightenment (17th and 18th centuries);
 - ▶ The political and cultural characteristics of countries;
 - ▶ Attributes such as hard work and saving;
 - ▶ The abundance of coal and access to colonies.
- ▶ **David Landes** (1924–2013), economic historian: institutions matter.
“Moreover, China lacked the institutions that fostered the cumulative process of discovery and learning: schools, universities, the diffusion of knowledge in society, challenges and competition.”

1. **What is economics?** → from Smith to CORE: interactions, livelihoods, change over time.
2. **The capitalist revolution** → the hockey stick of GDP per capita since 1700.
3. **The economist's tools** → a good question, and data + models + narratives.
4. **The Malthusian model** → diminishing returns + demographics: subsistence, 13th to 17th centuries.
5. **The technology-choice model** → iso-costs and the wage-capital ratio: dear labour, capital-intensive innovation, the great escape.
6. **Other related issues** → creative destruction, institutions, other factors.

Key concepts: GDP, ceteris paribus, production function, equilibrium, production technology, iso-cost line, wage-capital ratio.